University of BRISTOL

Lecture 8
Blade boundary conditions

ASDA/Dynamics of Rotor Systems
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This lecture

- Blade domain and its subdivision
- Boundary conditions for simple and advanced blade configurations
- Examples

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Single and multi-domain PDE cases

We introduced **IV-BVP** as being characterized by the space and time-dependent PDEs. These equations are also known as the field equations (FEs). The spatial domain of the definition *D* can consist of a single or multiple interconnected 1-dimensional (1D) subdomains. The end conditions for each of these subdomains are known as the boundary conditions (BCs).

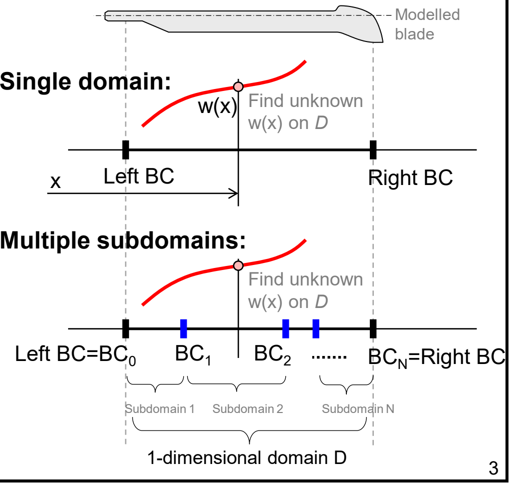
IV-BVP

FE

Field Equations (PDEs) describe internal behaviour of all subdomains in *D*.

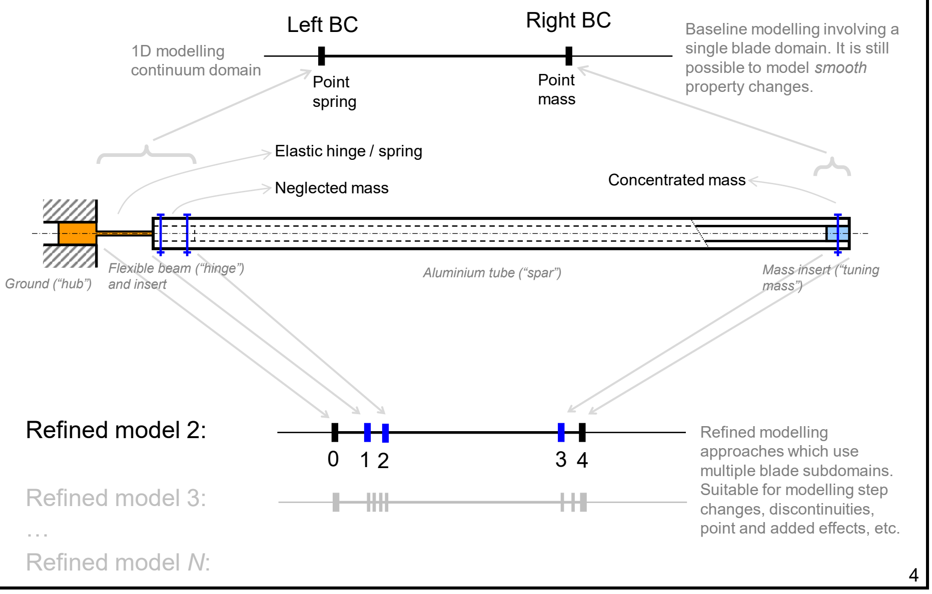
BC

Boundary Conditions (ODEs) describe the solution behaviour at external and internal boundaries using **geometric continuity** and **load-equilibrium** arguments.



Modelling fidelity decisions – lab example

Baseline model 1:



Main elastic blade equations so far …

Our models are represented by either single or multiple PDEs which describe one or multiple unknown functions of one spatial variable (“*x*”) and time (“*t*”). These PDEs are also called Field Equations (FEs). A sufficient number of Boundary Conditions (BCs) must be defined to find the unique solutions.

	No. of BCs per single domain	*No. of BCs per N domains*
Uncoupled bending and twist		
$\left(EI(x)\,w''(x)\right)'' = q(x)$	4	N×4
$(GJ(x)\,\theta'(x))' = -t(x)$	2	N×2
Uncoupled flapping and lead-lag		
$\left(EI_F\,w''\right)'' - \left(T\,w'\right)' + m\ddot{w} = L$	4	N×4
$\left(EI_L\,v''\right)'' - \left(T\,v'\right)' + m(\ddot{v} - \Omega^2 v) = D$	4	N×4
Coupled bending-twist		
$(EI\,w'')'' + m\,\ddot{w} + me\,\ddot{\theta} = L$	4	N×4
$(GJ\,\theta')' - I_{e.c.}\ddot{\theta} - me\,\ddot{w} = -(L\,e_A + M_A)$	2	N×2

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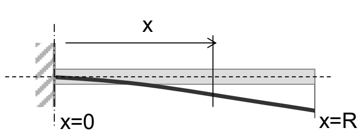
Elementary example – cantilever beam

Consider a stationary cantilevered beam with a free tip (e.g., non-rotating flapping blade). Write down all BCs for this case.

Flapping PDE when Ω=0 rad/s: $\left(EI_F\,w''\right)'' + m\,\ddot{w} = 0$

Other useful relationships: $M = EI\,w''$ $\quad S = Tw' - M' = -EI\,w'''$

Boundary conditions: (2 × root) + (2 × tip)



$$x = 0,\;\; w(0,t) = 0$$
$$x = 0,\;\; w'(0,t) = 0$$

Cantilever root – no motion (transversal displacement and rotation) permitted.

$$x = R,\;\; M(R,t) = 0 \Rightarrow w''(R,t) = 0$$
$$x = R,\;\; S(R,t) = 0 \Rightarrow w'''(R,t) = 0$$

Free tip implies vanishing loads of all types – in this case zero bending moment and transversal shear.

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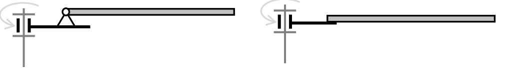
Usual blade-hub configurations

Evaluate all *known facts* about the boundary in question (blade root or tip) in terms of its mobility (displacement geometry) and loading (presence of external loading sources such as springs, masses, dampers, actuators, other subsystems, etc.). Boundary conditions represent constraints on the sought solutions such that the physics-based boundary requirements are satisfied whilst the solution continuity is maintained.



	Rotors	*Articulated*		*Hingeless*	
	Analytical relationships	Blade root $y=e_F R$	Blade tip $y=R$	Blade root $y=0$	Blade tip $y=R$
Displacement	$w(x,t)$	$w=0$	---	$w=0$	---
Slope	$w'(x,t)$	---	---	$w'=\beta_{built-in}$	---
Moment	$M=EI\,w''(x,t)$	$w''=0$	$w''=0$	---	$w''=0$
Shear	$S=T\,w'(x,t)-(EI\,w''(x,t))'$	---	$w'''=0$	---	$w'''=0$

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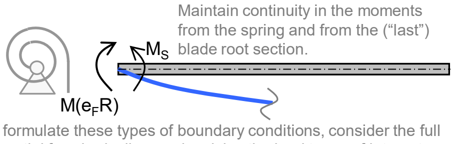
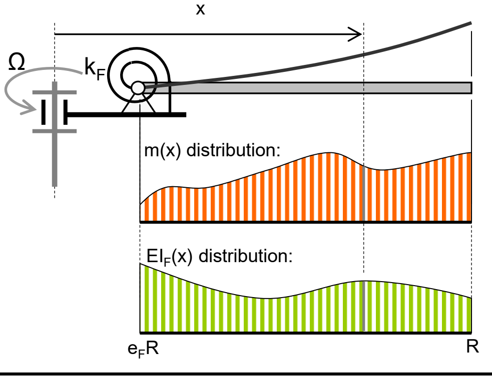
Advanced boundary conditions – root spring

Many rotor hub types, including articulated designs, have non-zero blade flapping (or lead-lag) hinge stiffness caused by effective or discrete elastic components (e.g., flexbeams, elastomeric joints). It is therefore important to consider this type of boundary condition.



To formulate these types of boundary conditions, consider the full or partial free-body diagram involving the load types of interest.



The root (torsional) spring BC is described by the equation of equilibrium between:

- internal moment $M(e_F R,t)=EI_F * w''$
- restoring moment $M_S=k_F * w'$

$$EI_F(e_F R)\,w(e_F R,t)''=k_F\,w(e_F R,t)'$$

$$\downarrow$$

$$w\,(e_F R,t)''-(k_F/EI_F)\,w\,(e_F R,t)'=0$$

$$\downarrow$$

$$\begin{bmatrix} w(e_F R,t) \\ w''(e_F R,t)-(k_F/EI_F)w'(e_F R,t) \\ w''(R,t) \\ w'''(R,t) \end{bmatrix}=\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

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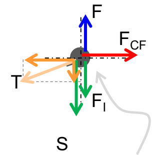
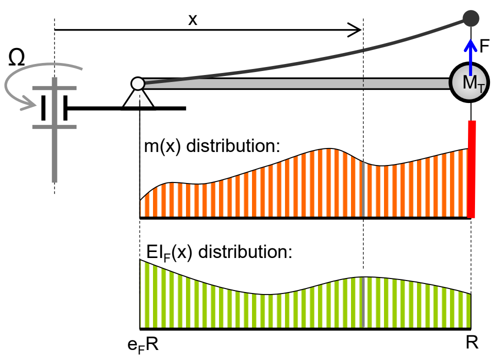
Advanced boundary conditions – tip mass

Many rotor blades have tip masses to ensure correct rotor balancing or dynamic blade tuning ("resonance avoidance"). It is therefore important to consider this type of boundary condition. This case also considers the externally applied point force.

A complete force FBD for the forces arising in the tip-mass segment. Both vertical (transversal) and axial (longitudinal) loads are considered. If the mass element had non-zero rotational inertia, additional consideration would be required to satisfy moment equilibrium.





The tip BC is constructed by considering the equation of equilibrium between:

- internal shear S(R,t)=… ,
- local inertial force $F_I(R,t)$=… ,
- applied force F(t).

$$M_T\ddot{w}(R,t)+\Omega^2 R M_T w'(R,t)$$
$$-EI_F(R)w'''(R,t)=F(t)$$

↓

$$\begin{bmatrix} w(e_F R,t) \\ w''(e_F R,t) \\ M_T\ddot{w}(R,t)+\Omega^2 R M_T w'(R,t)+EI_F(R)w'''(R,t)-F(t) \\ w''(R,t) \end{bmatrix}=\begin{bmatrix}0\\0\\0\\0\end{bmatrix}$$

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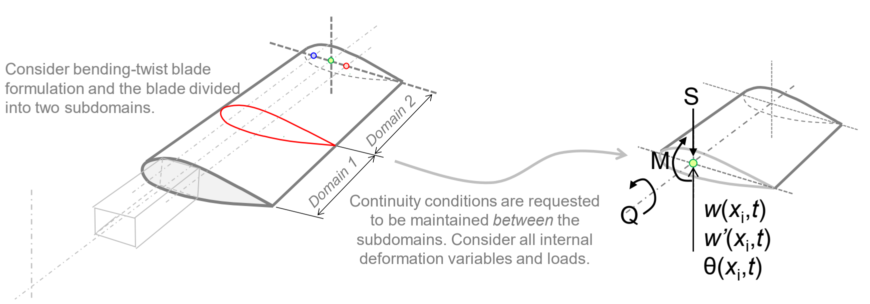
BCs for the bending-twist configuration

Consider now a blade with *abruptly spatially varying sectional properties* or *point loading effects* (balancing masses, devices such as pitch link, etc.). The points of "transition" or discontinuity (in the deformation geometry or internal loads) will require the *internal boundaries* between the interfacing subdomains. This approach leads to multi-point IV-BVPs.



For this case, each subdomain requires (4 × bending) + (2 × twist) BCs based on the *deformation continuity* (kinematic compatibility) and/or load equilibrium (dynamic compatibility). The load equilibrium conditions at the interface:

$$\sum_{(i)} Q_i=0, \sum_{(i)} M_i=0, \sum_{(i)} S_i=0$$

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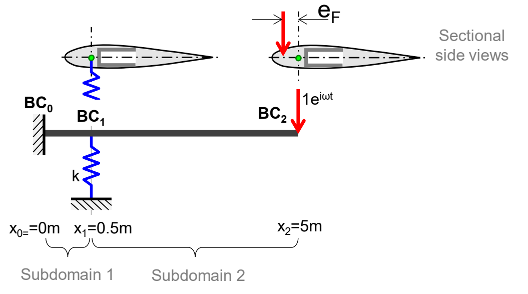
Advanced solved example

Consider a non-rotating blade in vacuum with the bending-twist coupling and a discrete spring. The blade is excited using harmonic load applied at the tip as shown. The line of spring action intersects the elastic axis (a line of sectional e.c. points) and the tip load $1e^{i\omega t}$ is applied vertically with an offset e_F.



Field equations:

$$(EI\,w'')'' + m\,\ddot{w} + me\,\ddot{\theta} = 0$$
$$(GJ\,\theta')' - I_{e.c.}\ddot{\theta} - me\,\ddot{w} = 0$$

Constitutive relationships:

$$M = EI\,w''$$
$$Q = GJ\,\theta'$$

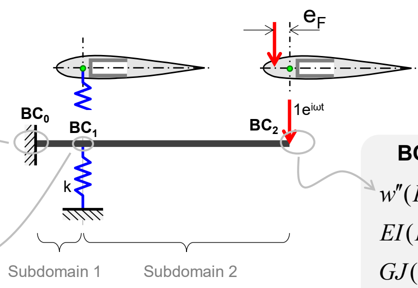
Balancing the order of the PDEs and the number of the required BCs: each subdomain requires 4+2 BCs. We have to formulate 12 BCs in total.

Advanced solved example, continued …

Boundary conditions for the two subdomains:

BC_0

$$w(0,t) = 0$$
$$w'(0,t) = 0$$
$$\theta(0,t) = 0$$



BC_2

$$w''(R,t) = 0$$
$$EI(R)\,w'''(R,t) = 1e^{i\omega t}$$
$$GJ(R)\,\theta'(R,t) = e_F(1e^{i\omega t})$$

BC_1

Geometric continuity

$$w(x_1^-,t) = w(x_1^+,t)$$
$$w'(x_1^-,t) = w'(x_1^+,t)$$
$$\theta(x_1^-,t) = \theta(x_1^+,t)$$

Local load equilibrium

$$EI(x_1^-)\,w''(x_1^-,t) = EI(x_1^+)\,w''(x_1^+,t)$$
$$EI(x_1^-)\,w'''(x_1^-,t) = EI(x_1^+)\,w'''(x_1^+,t) + kw(x_1^+,t)$$
$$GJ(x_1^-)\,\theta'(x_1^-,t) = GJ(x_1^+)\,\theta'(x_1^+,t)$$

Advanced solved example, continued …

Complete model – summary:

Field equations:

$$(EI\,w'')'' + m\,\ddot{w} + me\,\ddot{\theta} = 0$$

$$(GJ\,\theta')' - I_{e.c.}\ddot{\theta} - me\,\ddot{w} = 0$$

Boundary conditions:

$$\begin{bmatrix} w(0,t) \\ w'(0,t) \\ \theta(0,t) \\ w(x_1^-,t) - w(x_1^+,t) \\ w'(x_1^-,t) - w'(x_1^+,t) \\ \theta(x_1^-,t) - \theta(x_1^+,t) \\ EI(x_1^-)\,w''(x_1^-,t) - EI(x_1^+)\,w''(x_1^+,t) \\ EI(x_1^-)\,w'''(x_1^-,t) - EI(x_1^+)\,w'''(x_1^+,t) - kw(x_1^+,t) \\ GJ(x_1^-)\,\theta'(x_1^-,t) - GJ(x_1^+)\,\theta'(x_1^+,t) \\ w''(R,t) \\ EI(R)\,w'''(R,t) - 1e^{i\omega t} \\ GJ(R)\,\theta'(R,t) - e_F(1e^{i\omega t}) \end{bmatrix} = \mathbf{0}$$

Summary

- The total number of required boundary conditions depends on the overall order of the PDEs representing the problem and the number of solution subdomains.
- Boundary conditions are based on either kinematic continuity or dynamic equilibrium considerations.
- Different blade configurations and their integration in rotor hubs require specific sets of boundary conditions.
- External and internal boundaries, and their corresponding boundary conditions, can be used to model discontinuities in the model properties as well as other point effects such as loads and lumped parameter effects (e.g., dampers).
- External and internal boundaries, and their corresponding boundary conditions, can become the source of PDE coupling (e.g., bending-twist coupling).